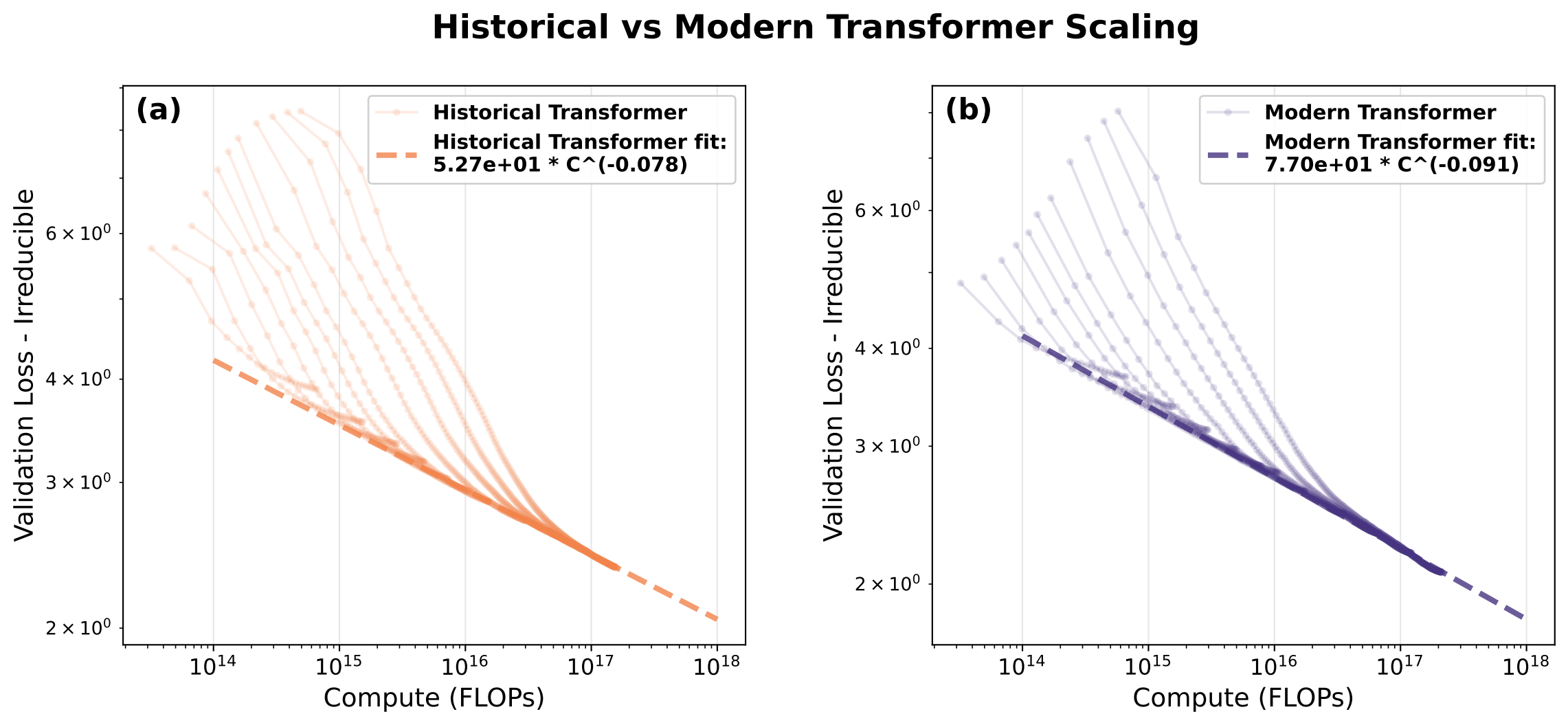

Historical vs Modern Transformer Scaling
(a)
Historical Transformer
Historical Transformer fit:
5.27e+01 * C^(-0.078)
Validation Loss - Irreducible
6 × 10^0
4 × 10^0
3 × 10^0
2 × 10^0
10^14
10^15
10^16
10^17
10^18
Compute (FLOPs)
(b)
Modern Transformer
Modern Transformer fit:
7.70e+01 * C^(-0.091)
Validation Loss - Irreducible
6 × 10^0
4 × 10^0
3 × 10^0
2 × 10^0
10^14
10^15
10^16
10^17
10^18
Compute (FLOPs)